CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
3rd Semester of B. Pharm. Examination
University Theory Examination Nov/Dec 2016
PH204 Pharmaceutical Analysis I

Date: 22.11.2016, Tuesday **Time:10:00 a.m to 1:00p.m** **Maximum Marks: 80**

Instructions:
1. **Figures to right indicate marks.**
2. **Section wise separate answer book to be used.**
3. **Draw neat sketches wherever necessary.**

SECTION- I

Q 1		Attempt any **TWO** of the following	
	A	[1] Define (a) pH, (b) Buffer, (c) Indicator, (d) Ionic product, (e) neutralization curve	**05**
		[2] Write a note on common ion effect.	**05**
	B	[1] Explain Buffer action with suitable examples.	**05**
		[2] Describe law of mass action.	**05**
	C	[1] Describe the different types of errors.	**05**
		[2] Write a note on error minimization.	**05**
Q 2		Attempt any **TWO** of the following	
	A	Define extraction. Explain Simple extraction, Multiple extraction and Continous extracton in brief.	**10**
	B	[1] Explain the different types of precipitation	**06**
		[2] Explain the two different types of nucleation.	**04**
	C	Elaborate the steps involved in gravimetric analysis of a compound.	**10**

SECTION- II

Q 3		Attempt any **TWO** of the following	
	A	[1] State the principle of non-aqueous titration.	**03**
		[2] Explain the leveling and differentiating effects seen in non-aqueous titration.	**07**
	B	[1] Enlist the ideal requirements for precipitation titration.	**05**
		[2] Explain the Mohr's Method.	**05**

C [1] Explain the Fajan's method of precipitation titration. **06**

[2] State the principle involved in Gasometry. **04**

Q 4 Attempt any **TWO** of the following

A Explain Iodometry and iodimetry, and state the difference between them. **10**

B [1] State the principle involved in Karl Fisher Titration. Give the significance of each reagent used in Karl fisher titration. **05**

[2] Enlist the different types of indicators used in redox titration and explain self indicators in brief. **03**

[3]Calculate the Redox Potential of the following cell **02**

$$Zn\ |Zn^{+2}\ ||\ Ag^{+1}\ |\ Ag$$

$$E^0_{\ Zn|Zn2+} = -0.76V \quad Zn \rightarrow Zn^{2+} + 2e^-$$

$$E^0_{\ Ag1+|Ag} = 0.337V \quad 2Ag^{1+} + 2e^- \rightarrow 2Ag$$

C [1] Justify: EDTA is commonly used for complexometric titrations. **04**

[2] Explain in brief the types of complexometric titrations. **06**